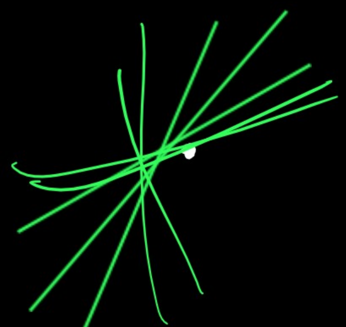
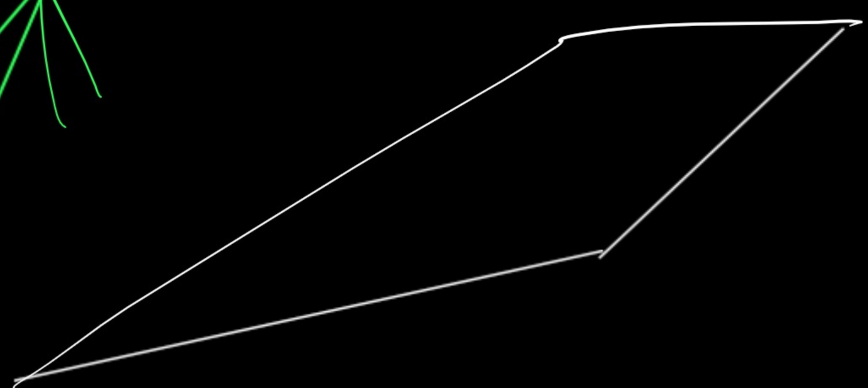
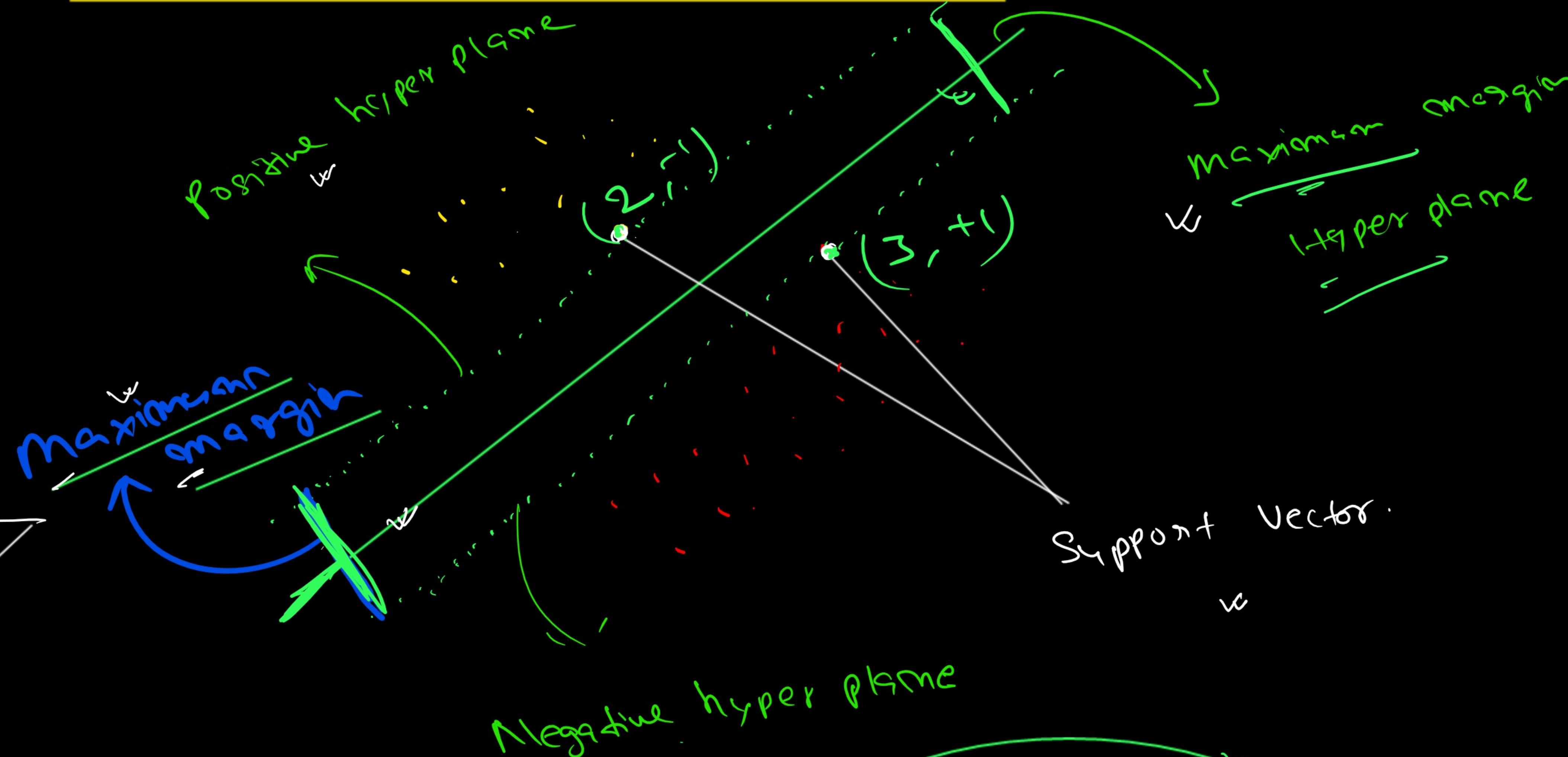
SVM → (Support Vector machine)

SVC, SVR

$\left(\frac{mm}{2}\right)$







(x, y) (x, y, z)

$$\text{max margin} = \frac{2}{\|w\|} \leftarrow \min_{(w,b)} \frac{1}{2}\|w\|^2$$

Point	x_i	y_i
A	1	-1
B	2	-1
C	3	+1
D	4	+1
E	5	9

$f(x) = \underline{w}x + \underline{b}$

$f(x) > 0 \rightarrow$ class +1

$f(x) < 0 \rightarrow$ class -1

Closest negative to positive is B at 2

" positive to negative is C at 3

$\text{boundary} = \dfrac{2+3}{2} = 2.5$

$(2, 3) \rightarrow$ Support vector

For B: $x = 2$, $y = -1$

$$f(2) = w \times 2 + b = -1$$

$w = 2$

$b = -5$

$$f(x) = 2x - 5$$

For C: $x = 3$, $y = +1$

$$f(3) = w \times 3 + b = +1$$

$A(1) \Rightarrow f(1) = 2 \times 1 - 5 = -3$ -ve

$B(2) \Rightarrow f(2) = 2 \times 2 - 5 = -1$ -ve

$C(3) \Rightarrow f(3) = 2 \times 3 - 5 = +1$ +ve

$D(4) \Rightarrow f(4) = 2 \times 4 - 5 = +3$ +ve every training point (x_i, y_i)

$$y_i(w \cdot x_i + b) \rightarrow \geq 1$$

Lagrangian $\Rightarrow$ ~~Objective~~ $- \sum$ (multiplier $\times$ constraint)

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum \alpha_i \left[y_i (w \cdot x_i + b) - 1 \right]$$

$$\frac{\partial L}{\partial w} = 0 \qquad \frac{\partial L}{\partial b} = 0$$

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{N} \alpha_i y_i x_i$$

$$\underline{w} = \sum_{i=1}^{N} \alpha_i y_i x_i$$

$$\frac{\partial L}{\partial \alpha_i} = -\left[y_i (w \cdot x_i + b) - 1 \right]$$

$$y_i (w \cdot x_i + b) \geq 1$$

$$\frac{\partial L}{\partial b} = - \sum_{i=1}^{N} \alpha_i y_i$$

$$\sum_{i=1}^{N} \alpha_i y_i = 0$$

Dual objective function

$$\max_{\alpha} \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

Confusion matrix

		+1	−1
	y \ $\hat{y}$	Predicted Disease	Predicted Healthy
+1 →	Actual Disease	40 TP	10 FN
−1 →	Actual Healthy	5 FP	45 TN

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN}$$

$$\text{F1 Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$